SmartPark: Predicting Parking Availability Using Machine Learning

Kerri Jensen[1], Rebekah Jensen[2] , Dr. Prosenjit Chatterjee [3]
[1]Department of Computer Science and Cybersecurity (under CSTEM), [2] Southern Utah University

SUU SOUTHERN UTAH UNIVERSITY

Introduction

Drivers waste time searching for available parking, contributing to congestion and user frustration. While many smart parking systems rely on physical infrastructure, computer vision offers a scalable alternative using image data alone [1-4]. This project evaluates progressively deeper machine learning approaches for parking analysis. SmartPark investigates what level of modeling is necessary to move from simple occupancy detection to fine-grained parking availability estimation.

Research Goal

To evaluate machine learning approaches for parking occupancy estimation and determine what level of modeling complexity is required for fine-grained availability prediction.

Methods

PKLot - Dataset (Parking Lot - PK)
Public aerial parking dataset, full-resolution parking images
Frame-level labels: EMPTY (0) / OCCUPIED (1)

Random Forest Regressor (RFR) - Temporal Prediction
Hour, weekday, weekend indicator (simulated)
Output: Predicted number of available spots

Convolutional Neural Network (CNN) - Occupancy Classification
Output: EMPTY (0) / OCCUPIED (1)

YOLOv8 - Vehicle Localization
Output: Vehicle bounding boxes and counts

Figures and Results

```
MAE: 5.17347719205631
R^2: 0.4663569050282075
```

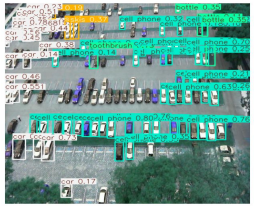
RFR Results

Trained on simulated temporal features (hour, weekday, weekend)
Interpretation: Basic temporal patterns explain ~47% of availability variance
Limitation: Performance may differ with real-world occupancy data and additional contextual features

Figures and Results (cont.)



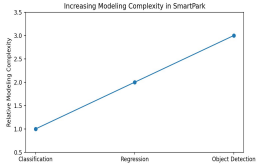
CNN run shows lot is occupied



Initial run Vehicles detected: 24



3rd run Vehicles detected: 14



CNN Results

Accuracy: 1.00 Recall: 1.00
Precision: 1.00 F1-Score: 1.00
Trained 3 epochs on PKLot training split

```
Confusion Matrix:
[[  26    0]
 [   0 1216]]
```

Interpretation: Learns dominant signal quickly due to strong class imbalance and clear visuals
Limitation: Dataset highly imbalanced (~0.98% interpreted as occupied); frame-level labels cannot identify individual vacant spots

YOLOv8 Results

Pre-trained YOLOv8 given COCO (Common Objects in Context) weights with vehicle classes filters (car, bus, truck, motorcycle). Not trained on YOLOv8, but could be implemented.
Interpretation: Successfully localizes vehicles in some images but shows inconsistency across aerial perspectives. Mislabels. Pre-trained, so quality of results is randomized.
Limitation: PKLot lacks bounding box annotations, preventing parking-specific fine-tuning.

Conclusion

This study evaluated progressively deeper machine learning approaches for parking analysis. A CNN achieved near-perfect frame-level occupancy classification on the PKLot dataset, while a Random Forest Regressor demonstrated that temporal features explain nearly half of parking availability variance.

Pre-trained YOLOv8 successfully localized vehicles in some aerial images; however, it was originally trained on the COCO dataset rather than parking-specific imagery. This domain mismatch, combined with the absence of bounding-box annotations in PKLot, limited fine-grained availability estimation.

Overall, results suggest that the primary limitation in parking intelligence is not model capability, but dataset structure. Future work should focus on creating localization-ready parking datasets to enable slot-level availability prediction.



Labels are not necessary. Bounding boxes represent location of vehicles.

Acknowledgments

We thank Dr. Prosenjit Chatterjee and Southern Utah University for their support and guidance throughout this research.

References

[1] Acharya, D., Yan, W., & Khoshelham, K. (2018). Real-time image-based parking occupancy detection using deep learning.
[2] Almeida, P. R., Oliveira, L. S., Silva Jr, E., Britto Jr, A. S., & Koerich, A. L. (2015). PKLot – A robust dataset for parking lot classification. Expert Systems with Applications, 42(11), 4937–4949.
[3] Awan, F. M., Saleem, Y., Minerva, R., & Crespi, N. (2020). A comparative analysis of machine/deep learning models for parking space availability prediction. Sensors, 20(1), 322. https://doi.org/10.3390/s20010322
[4] Jocher, G., Qiu, J., & Chaurasia, A. (2023). Ultralytics YOLO (Version 8.0.0) [Computer software]. https://github.com/ultralytics/ultralytics